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MTH 302 — Past Questions & Materials
by Renance

Course: MTH 302
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — MTH 302 Or 312 Ordinary Differential Equations 2018 2019 first semester

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
DEPARTMENT OF MATHEMATICS/COMPUTER SCIENCE.
FIRST SEMESTER EXAMINATION 2018/2019 SESSION.
MTH 312: Ordinary Differential Equations I.

Instructions: Answer any four questions. Time $2\frac{1}{2}$ hrs

1. a. Let y_1 and y_2 be two solutions of $y'' + q(x)y' + r(x)y = 0$
i. Define the Wronskian of y_1 and y_2 .
ii. State the necessary condition for the linear independence of y_1 and y_2 .

Using method of undetermined coefficients, write down the form of the particular solution to:

- b. Determine if the following sets of functions are linearly dependent or linearly independent.
i. $f(t) = \cos t$, $g(t) = \sin t$
ii. $f(y) = e^{\lambda y} \cos(\mu y)$, $g(y) = e^{\lambda y} \sin(\mu y)$
iii. $f(x) = 6^x$, $g(x) = 6^{x+2}$

2. a. Find the general solution to $2t^2y'' + ty' - 3y = 0$, $t > 0$
given that $y_1(t) = t^{-1}$ is a solution.
b. Find the general solution to $t^2y'' + 2ty' - 2y = 0$, $t > 0$
given that $y_1(t) = t$ is a solution.

3. a. Suppose $Y_1(t)$ and $Y_2(t)$ are two solutions to the ODE

$$y'' + p(t)y' + q(t)y = g(t)$$

and that $y_1(t)$ and $y_2(t)$ are a fundamental set of solutions to the associated homogeneous differential equation

$$y'' + p(t)y' + q(t)y = 0$$

Show that $Y_1(t) - Y_2(t)$ is a solution to the non homogeneous ODE and can be written as $Y_1(t) - Y_2(t) = c_1y_1(t) + c_2y_2(t)$

- b. Using variation of parameters, find the general solution to the following differential equation

$$y'' - 2y' = \frac{e^t}{t^2 + 1}$$

4. a. Define the following
i. Analytic Function